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AN EMPIRICAL MODEL OF COLLECTIVE HOUSEHOLD LABOUR SUPPLY WITH NON-PARTICIPATION*

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I present a structural empirical model of collective household labour supply that includes the non-participation decision. I specify a simultaneous model for hours, participation and wages of husband and wife. I discuss the problems of identification and statistical coherency that arise in the application of the collective household labour supply model. The model includes random effects and it is estimated using a panel data set of Dutch couples. The estimates allow me to check the underlying regularity conditions on individual preferences and to obtain estimates of the sharing rule that governs the division of household income between husband and wife.

The literature on collective models of household labour supply is growing. Traditionally, the unitary model has been used to explain the labour supply behaviour of households. The unitary model is the direct application of the neoclassical labour supply model for *individual* labour supply behaviour to the household as a decision unit. This approach has been criticised for several reasons. One of the fundaments of micro-economic theory is that the individual is the decision making unit. The unitary model assumes that a utility function exists for the household as a whole and leaves the underlying preferences of the household members unspecified. An implication of the unitary model is income pooling, which means that the explanation of working hours of household members is based on the pooled household income, while the source of the income components does not matter. This implication places the income pooling restrictions on the labour supply functions of individual household members, which are often rejected in empirical work. Furthermore, the unitary model implies that the Slutsky conditions hold. Empirical studies often reject the Slutsky conditions for households with more than one member.

These considerations have led to the development of models of household decision making that allow for individual preferences and bargaining between household members. McElroy and Horney (1981) introduced Nash bargaining in a household decision model. Apps and Rees (1988) do not impose a specific bargaining rule but assume efficiency. The collective model of household labour supply is explicitly based on individual preferences of the household members. Chiappori (1988; 1992) shows that under the assumption of Pareto efficiency of the household decision process and the assumption that individual preferences are of the *egoistic* or *caring* type it is possible to derive testable implications for the labour supply behaviour of individual household members.

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The household decision process can be represented by a two-stage budgeting process:[1] in the first stage the non-labour income of household members is divided between the members and in the second stage, conditional on this division, individuals make their labour supply decision. Chiappori (1988) shows that under the assumptions mentioned it is possible to recover the sharing rule of non-labour income between household members non-parametrically, up to an additive constant.

Fortin and Lacroix (1997) empirically analyse the collective household labour supply model. For a Canadian cross-section dataset of two-earner households, they estimate a collective household labour supply model with a flexible functional form. They carry out a complete set of tests of restrictions implied by the collective model and by the unitary model. However, as opposed to what is usual in the labour supply literature since the work by Heckman (1979), the specification of Fortin and Lacroix (1997) does not incorporate non-participation and corner solutions. The reason for this is that the modelling of the participation decision in the collective model is much more complicated than in the unitary model. Since the wage rate enters the labour supply function both directly, as in the unitary model and, indirectly, through the sharing rule, the derivation of a unique reservation wage is non-trivial and requires the placement of additional restrictions on the preferences of household members. However, for empirical work on collective household labour supply the inclusion of non-participants is very important since restricting the sample to couples with working spouses may lead to selectivity bias in the estimates of the labour supply functions and the sharing rule.

Recently, the inclusion of non-participants in the collective labour supply model has been dealt with theoretically, both by Donni (2003) and by Blundell *et al.* (2007). Donni (2003) leaves the hours choice of husband and wife unrestricted: both household members can choose not to work or to work any positive amount of working hours. Blundell *et al.* (2007) assume that men can only choose to work 40 hours a week or choose not to work at all. Donni (2003) shows that the identification result by Chiappori (1988) for the sharing rule can be extended to the case of a non-participating household member.

In this study I expand on the empirical literature of collective household labour supply. I specify a model that allows for non-participation. My model includes information on working hours of both husband and wife to estimate the labour supply functions for both spouses. A serious complication that arises in modelling both the hours and the participation decision is the problem of statistical coherence. I derive parameter restrictions that are sufficient for the coherence of my model.

The model that I specify contains equations for the working hours of husband and wife and for the wages of both spouses. In the econometric specification I allow for the endogeneity of wage rates and for correlation between working hours of husband and

[1] An underlying assumption for this two-stage procedure is that consumption of household members can be represented by a Hicks aggregate commodity and that there is no public good within the household. Children are often mentioned as an example of a public good that affects the division of time between household members. As a consequence, in empirical studies of collective household labour supply the sample is either restricted to couples without children (Blundell *et al*, 2007), or to couples without young children (Fortin and Lacroix, 1997).

wife due to unobserved heterogeneity. In the estimation, I use simulation methods to integrate over unobserved wage rates: I estimate the parameters of the hours and participation equations and the wage distribution simultaneously by simulated maximum likelihood (SML).

Estimation of the model gives estimates of the sharing rule and of the labour supply functions of husband and wife. Since the marital status may influence the relative bargaining position of husband and wife, I follow an estimation strategy that allows the parameters of the sharing rule to be fully flexible by marital status. Using a minimum distance estimation criterion, I test for equality of the labour supply and sharing rule parameters by marital status.

I use data from the Dutch Socio Economic Panel for the years 1990–2001. This is a panel survey among Dutch households. I select a panel data set of Dutch couples who are living without children. The data contain detailed information about the labour market status, working hours, wage rates and various background characteristics of both husband and wife. I exploit the panel feature of the data by allowing for random effects in the labour supply equations and the wage equations.

Section 1 describes the collective model of labour supply which provides the necessary background for my model. Section 2 specifies the empirical model. Here I address problems of identification and statistical coherence. Section 3 contains a description of the available data. Section 4 presents the results of estimation. Section 5 concludes.

1. The Collective Model: Theoretical Framework

This Section describes the collective model of labour supply. I formulate the model and discuss the identification results of the sharing rule, as derived by Chiappori (1988; 1992), and the extended results that allow for corner solutions by Blundell *et al.* (2007) and Donni (2003).

In what follows I consider the labour supply decision of a household, consisting of husband and wife. Working hours, consumption and the wage rate of the husband and wife are denoted by (h_m, C_m, w_m) and (h_f, C_f, w_f), respectively, while y denotes the household's non-labour income. The identification results of the sharing rule require household preferences to be of the egoistic (or caring) type. Moreover, the sharing rule representation of the collective labour supply problem requires the absence of public goods within the household.[2]

[2] In reality, there are many potential public goods in a household. But within the literature on collective labour supply models, children are considered as collective goods that have a potentially large influence on the division of time between household members. For this reason the analysis is usually restricted to childless couples; see e.g. Fortin and Lacroix (1997). Recently, Chiappori *et al.* (2005) extended the collective model to allow for public goods. However, an empirical application of this approach to children in the household requires the availability of information on household expenditures on children. Another problem is that children are related to domestic production. In empirical labour supply models non-work time is usually interpreted as leisure time, ignoring domestic production. In a survey article Apps (2003) discusses the problems that arise in empirical labour supply models due to neglecting the domestic production and alternative time uses.

In this Section I assume preferences are egoistic.[3] The utility level of household member i, working h_i hours and consuming C_i, is denoted by $u^i(1 - h_i, C_i)$. I assume that the utility function satisfies the usual regularity conditions.

The collective approach assumes that households only take Pareto efficient decisions. Formally, for any combination of wages and non-labour income (w_m, w_f, y) a utility level $\bar{u}^m(w_m, w_f, y)$ exists such that (h_m, C_m, h_f, C_f) is the solution to the following decision problem:[4]

$$
\begin{aligned}
&\max_{h_m, C_m, h_f, C_f} u^f(1 - h_f, C_f),\\
&u^m(1 - h_m, C_m) \geq \bar{u}^m(w_m, w_f, \mathrm{y}),\\
&C_m + C_f = w_m h_m + w_f h_f + y,\\
&0 \leq h_m, h_f \leq 1.
\end{aligned} \tag{1}
$$

The utility level $\bar{u}^m(w_m, w_f, y)$ can be interpreted as the outcome of a bargaining process between husband and wife, and the wage rates of husband and wife may influence the bargaining power of the household members. The bargaining process itself is left unspecified. An alternative representation of (1) follows from applying the second fundamental theorem of welfare economics. A pair of labour supply functions (h_m^*, h_f^*) is collectively rational (i.e. satisfies (1)) if a combination $(\bar{C}^m, \bar{C}^f)$ and a pair of functions (ρ^m, ρ^f) exists such that $(h_i^*, \bar{C}_i)$ is a solution to

$$
\begin{aligned}
&\max_{h_i, C_i} u^i(1 - h_i, C_i),\\
&\text{subject to } C_i = w_i h_i + \rho^i, i = m, f,\\
&\text{with } \rho^m + \rho^f = y.
\end{aligned} \tag{2}
$$

Thus, the collective rational household labour supply decision can be represented by a two-stage budgeting problem. In the first stage the share $\rho^i, i = m, f$, of non-labour income y of each household member is determined. The share depends on the bargaining power of the household members, which in turn depends on the wages of the members and the non-labour income, and I denote $\rho^m = \rho(w_m, w_f, y)$ and $\rho^f = y - \rho(w_m, w_f, y)$.[5] Note that ρ is not restricted to be positive so transfers from one household member to the other are possible.

The decision problem (2) is the basis for deriving the restrictions that the collective approach imposes on the labour supply function. Let $H^i(w_i, \rho^i)$ denote the 'structural' labour supply function conditional on ρ^i that follows from solving (2), and let $h_i^*(w_m, w_f, y)$ denote the 'reduced form' labour supply function. Then it follows that

[3] The assumption of egoistic, or caring, preferences is used by Chiappori (1988; 1992) to identify preference parameters from sharing rule parameters. The assumption of egoistic preferences places restrictions on the behaviour of households. For instance, it excludes the possibility of complementarities in leisure between spouses, or substitution in non-market time of spouses.

[4] Various alternative representations of the decision problem are possible. For example, the objective function can be written as a weighted average of the utility functions of husband and wife, with weights that depend on (w_m, w_f, y). This can directly be seen by applying Kuhn-Tucker to (1). See Vermeulen (2002) for an overview.

[5] This formulation relates the shares of spouses to aggregate household non-labour income. It excludes the possibility that shares of household non-labour income are assigned to either spouse on basis of ownership.

$$\begin{aligned} h_m^*(w_m, w_f, y) &= H^m[w_m, \rho(w_m, w_f, y)], \\ h_f^*(w_m, w_f, y) &= H^f[w_f, y - \rho(w_m, w_f, y)]. \end{aligned} \tag{3}$$

Thus, from (3) it follows that the wage rate of the other household member enters the labour supply function through the sharing rule.

The relationship in (3) is the basis for the identification result of Chiappori (1988; 1992) which reads that, given information on working hours, wages and non-labour income of both household members, in the absence of corner solutions, the sharing rule $\rho(w_m, w_f, y)$ can be recovered up to an additive constant. This means that it is possible to recover marginal effects of the respective wage rates of husband and wife and of the household's non-labour income on the sharing rule.[6] The sharing rule can be recovered by taking first and second order partial differentials of (3) which constitutes a system of differential equations from which the sharing rule can be solved.

The approach described so far does not account for the presence of corner solutions. In empirical models of collective labour supply corner solutions are usually ignored by restricting the sample to working couples. In contrast, in empirical models of individual labour supply the option of choosing zero working hours usually is accounted for to prevent selectivity bias. In models of individual labour supply, the reservation wage follows naturally from the model as the wage rate at which optimal working hours are exactly equal to zero, or, equivalently, as the wage rate that is equal to the marginal rate of substitution between consumption and labour supply, evaluated at zero working hours. In collective models of labour supply the reservation wage does not follow naturally from the model as the wage of the male affects male labour supply both directly, in the same way as in individual models of labour supply and, indirectly, through the sharing rule and, moreover, it affects female labour supply by the sharing rule. Both Donni (2003) and Blundell *et al.* (2007) address this problem in a different way. Donni (2003) takes as point of departure the model outlined above. This assumes that working hours of husband and wife both vary continuously. Blundell *et al.* (2007) assume that the hours of the wife vary continuously, whereas the weekly working hours of the husband are restricted to either zero or forty. Since my empirical model is based on the framework by Donni (2003), I will give an outline of the result by Donni (2003).

First Donni (2003) defines the function $\omega^i(w_m, w_f, y)$ by the marginal rate of substitution between leisure and consumption of household member i:

$$\omega^i(w_m, w_f, y) = -\frac{u_h^i[1, \rho^i(w_m, w_f, y)]}{u_c^i[1, \rho^i(w_m, w_f, y)]}, i = m, f. \tag{4}$$

In (4) the subscripts h and c indicate partial derivatives with respect to labour supply and consumption, respectively. Now the reservation wage rate for household member i is defined as the wage w_i for which $w_i = \omega^i(w_m, w_f, y)$. An additional assumption needs to be made to ensure uniqueness of the reservation wage rate. To this purpose Donni (2003) formulates

[6] This possibility for recovering the marginal effects depends on the assumption of egoistic, or caring, preferences. As discussed before, this assumption imposes restrictions on the preferences of spouses.

ASSUMPTION R1. *For any* (w_m^*, w_f^*, y) *and* $(\tilde{w}_m, \tilde{w}_f, y)$ *preferences and the sharing rule are such that*

$$\max_{i=m,f}\left[|\omega^i(w_m^*, w_f^*, y) - \omega^i(\tilde{w}_m, \tilde{w}_f, y)|\right] \leq \max_{i=m,f}(|w_i^* - \tilde{w}_i|). \tag{5}$$

This condition implies that the system of equations (4) is a contraction mapping with respect to the variables w_m and w_f. Intuitively, (5) requires that the impact of w_m and w_f on the shares ρ^m and ρ^f should not be 'too large'.

Condition (5) ensures uniqueness of the reservation wage in two senses. First, a unique pair of reservation wages $\hat{w}_m(y)$ and $\hat{w}_f(y)$ exists such that at wage rates w_i with $w_i > \hat{w}_i(y), i = m, f$ both household members prefer to work. Second, for household member i there exists a reservation wage $\gamma^i(w_j, y)$, given the wage w_j of the other household member, such that household member i prefers to work if $w_i > \gamma^i(w_j, y)$.

Both Donni (2003) and Blundell *et al.* (2007) provide a proof of the identification of the sharing rule (up to an additive constant) by extending the identification result by Chiappori (1988). The original identification result for the case in which both husband and wife participate is based on the labour supply equations in (3). Taking first and second order derivatives of (3) results in a system of partial differential equations, from which the sharing rule can be recovered up to an additive constant. Suppose that household member j works at wage w_j while household member i does not. In this situation, there is only one partial differential equation, namely the equation for the participating spouse j to identify the sharing rule from. Define the participation frontier of household member i as $w_i = \gamma^i(w_j, y)$ for $i,j = m, f, i \neq j$. Donni (2003) shows that the sharing rule can be identified from this differential equation if w_i approaches the participation frontier $\gamma^i(w_j, y)$.

2. Specification

2.1. *Preferences, Labour Supply and the Sharing Rule*

For the empirical implementation of the model, I specify the following labour supply function:

$$H^j(w_j, y_j) = \alpha_1^j + \alpha_2^j y_j + \alpha_3^j w_j + \alpha_4^j w_j^2, j = m, f. \tag{6}$$

The labour supply function (6) is flexible enough to allow for backward bending as it is quadratic in the wage rate.[7]

Hausman and Ruud (1984) showed that the underlying utility function of the labour supply function (6) is

$$u^j(h, c) = m^* \exp\left[\frac{\beta_j}{\gamma_j}(h - \delta_j - \beta_j m^*)\right], \tag{7}$$

with

[7] A similar specification is used in Bloemen and Kapteyn (2008) in an analysis of female labour supply and by Kapteyn *et al.* (1990) in the context of a household labour supply model. The latter study shows evidence of a backward bending labour supply function for men.

$$m^* = \frac{\gamma_j}{\beta_j^2}\left(1 - \left\{1 + \frac{\beta_j^2}{\gamma}\left[\frac{(h-\delta_j)^2}{\gamma_j} - 2(c + \vartheta_j)\right]\right\}^{\frac{1}{2}}\right), \tag{8}$$

for household member j, $j = m, f$. The parameters of (7), (8) and (6) are related by $\alpha_1^j = \delta_j + \beta_j\vartheta_j$; $\alpha_2^j = \beta_j$; $\alpha_3^j = \gamma_j + \beta_j\delta_j$; $\alpha_4^j = \beta_j\gamma_j/2$. Note that $\beta_j < 0$ and $\gamma_j > 0$ imply that $\alpha_2^j < 0$ and $\alpha_4^j < 0$.

Now I assume that the sharing rule $\rho(w_m, w_f, y)$ has the following form:[8]

$$\rho(w_m, w_f, y) = k_1 + k_2 w_m + k_3 w_f + k_4 \frac{w_m}{w_m + w_f} + k_5 w_m^2 + k_6 w_f^2 + k_7 y. \tag{9}$$

The sharing rule is specified as a quadratic function of the wage rates of both spouses and of the household's non-labour income. In empirical applications of the collective model, often a measure for the relative wage rates of the spouses is included to represent their relative bargaining power. Therefore the sharing rule includes the husband's wage rate expressed as a fraction of the sum of the wage rates of both spouses.[9]

Marital status may influence the relative bargaining position of husband and wife. In the empirical analysis, the parameters of the sharing rule (9) are allowed to vary with marital status b, and I may write $\kappa_j = \kappa_j(b)$, $j = 1,\ldots,7$. Hence, the relative bargaining position of partners within married and unmarried couples can be different. In Section 2 I discuss the estimation strategy to allow for variation in the sharing rule by marital status. For the moment it is sufficient to realise that if the sharing rule parameters vary with martial status, the parameters of the resulting labour supply function vary with marital status as well. In the remainder of the Section I suppress the dependence of the parameters on marital status in my notation. In my analysis I treat marital status as an explanatory variable. Becker (1973) and Manser and Brown (1980) explicitly model marriage, or couple formation, as a choice variable that is closely related to issues as time allocation and specialisation in home production by household members.[10]

[8] Note that before I have specified the share of the husband as $\rho^m = \rho(w_m, w_f, y)$ and the share of the wife as $\rho^f = y - \rho(w_m, w_f, y)$. Thus, additivity of the husband's and wife's share to the household's non-labour income y is imposed.

[9] I have experimented with a sharing rule with the higher order term $w_m/(w_m + w_f)y$ but this additional term turned out not to be significant. Browning *et al.* (1994) first recognised the importance of including some measure of the relative earning capacity of husband and wife. They discuss the identification of the sharing rule and stress that identification may depend on the non-linear shape the measure. I have done a sensitivity analysis with an alternative measure of relative wage rates of the spouses, the logarithm of wage rate ration of husband and wife. Both qualitative and quantitative outcomes were hardly affected by this choice.

[10] Manser and Brown (1980) state that the utility that each spouse within a couple derives from his or her consumption and leisure, should be at least as large as the utility he or she can obtain as a single. Becker (1973) argued that the gains from marriage come from the production of home produced goods with a production technology that exhibits increasing returns to time inputs, such that it can be beneficial to have specialisation by spouses in different market or non-market activities. The analysis of Manser and Brown (1980) also shows that it is empirically hard to test the model implications as wage rates of individuals both affect the utility of being single and influence the bargaining position of spouses within a couple. In my analysis, I only include marital status as an explanatory variable and thus I ignore the issue of formation of marriages. Keeley (1979) uses a search framework to describe the process of marriage formation. From the analysis by Manser and Brown (1980), it follows that marital status potentially may enter the analysis by two routes: it may influence the bargaining position of spouses within households and there may be heterogeneity in preferences that define the single individual threat point. Both may determine the marital status of a couple.

Inserting the sharing rule (9) in the 'structural' labour supply functions (6) (with $y_m = \rho$ and $y_f = y - \rho$), results in the 'reduced form' labour supply functions:

$$\begin{aligned} h_m(w_m, w_f, y) &= a_1 + a_2 w_m + a_3 w_f + a_4 \frac{w_m}{w_m + w_f} + a_5 w_m^2 + a_6 w_f^2 + a_7 y, \\ h_f(w_m, w_f, y) &= b_1 + b_2 w_m + b_3 w_f + b_4 \frac{w_m}{w_m + w_f} + b_5 w_m^2 + b_6 w_f^2 + b_7 y. \end{aligned} \tag{10}$$

The parameters of the sharing rule k_l, $l = 2,\ldots,7$ can be identified from the parameters of the labour supply functions (10):

$$\begin{aligned} &k_2 = b_2 a_4/\Delta, k_3 = b_4 a_3/\Delta, k_4 = b_4 a_4/\Delta, k_5 = b_5 a_4/\Delta, \\ &k_6 = b_4 a_6/\Delta, k_7 = b_4 a_7/\Delta, k_4 = b_4 a_4/\Delta, \Delta = b_4 a_7 - a_4 b_7. \end{aligned} \tag{11}$$

The parameters of the structural labour supply function (6) can be expressed in terms of the parameters in (10) as

$$\begin{aligned} &\alpha_2^m = \Delta/b_4; \alpha_3^m = a_2 - b_2 a_4/b_4; \alpha_4^m = a_5 - b_5 a_4/b_4; \\ &\alpha_2^f = -\Delta/a_4; \alpha_3^f = b_3 - b_4 a_3/a_4; \alpha_4^f = b_6 - b_4 a_6/a_4. \end{aligned} \tag{12}$$

A requirement for the identification of the parameters of the sharing rule is that $\Delta \neq 0$. From the restrictions above it follows that if $\Delta = 0$ then $\alpha_2^m = \alpha_2^f = 0$ and the sharing rule would not even enter the labour supply equations.

Note that the 'reduced form' labour supply functions (10) may be consistent with more general specifications of preferences. However, the possibility of recovering the sharing rule parameters (11) separately from the preference parameters (12) stems from the assumption of egoistic preferences and from the assumption that the wage rate fraction enters the labour supply equations by the sharing rule.

Leisure is a normal good for both husband and wife if $\alpha_2^j < 0, j = m, f$. A necessary condition for this is that the parameters a_4 and b_4 in the 'reduced form' labour supply functions (10) have the opposite sign, as can be seen from (12). From (12) it can be seen that the conditions $\alpha_2^j < 0, j = m, f$ are not trivial and therefore constitute a test of the theory.

Note that it is potentially important to allow for backward bending of the 'structural' labour supply functions. For instance, an increase in the husband's wage rate has a direct effect on his supply of labour, by the coefficients of the labour supply function (6) and an indirect effect, that runs through the sharing rule (9). Not allowing for backward bending will bias the estimates of the effect that runs through the sharing rule.

For future reference I introduce the vector notation $\phi_m = (a_1, a_2, a_3, a_4, a_5, a_6, a_7)'$, $\boldsymbol{\phi}_f = (b_1, b_2, b_3, b_4, b_5, b_6, b_7)'$ and $\mathbf{W} = [1, w_m, w_f, w_m/(w_m + w_f), w_m^2, w_f^2, y]'$.

The relations (11) between the labour supply parameters in (10) and the parameters of the sharing rule (9) hold for the case in which both husband and wife are working and can choose any number of working hours. The analysis by Donni (2003) shows how the labour supply model can be extended to the case in which the husband works and the wife does not. Donni (2003) proposes a switching regression model for labour supply of the husband. If the wife does not work, male labour supply becomes:

$$h_m(w_m, w_f, y) = \mathbf{\Phi}'_m\mathbf{W}. \tag{13}$$

The underlying reason for this 'switch' is a difference in the sharing rule due to the different female labour market state. If the wife works, her wage income actually affects the total household income and consequently influences the sharing rule. Apart from that, the wife's wage rate affects the sharing rule due to her bargaining power. If the wife does not work, only the effect of the wife's (potential) wage rate on the bargaining power remains.

As shown by Donni (2003) the parameters $\boldsymbol{\phi}_m$ and $\mathbf{\Phi}_m$ of male labour supply in case of a working and a non-working wife must satisfy restrictions for the labour supply function to be continuous along the participation frontier of the wife. This continuity holds for the following relation:

$$\mathbf{\Phi}'_m\mathbf{W} = \boldsymbol{\phi}'_m\mathbf{W} + s_m(\boldsymbol{\phi}'_f\mathbf{W}). \tag{14}$$

In (14) s_m is a parameter. Restriction (14) implies that if the wife is on the participation frontier (which means that female labour supply $\boldsymbol{\phi}'_m\mathbf{W}$ is equal to zero) the two regressions coincide. Blundell *et al.* (2007) give a more fundamental explanation of this continuity restriction postulated by Donni (2003). They introduce the 'double indifference' assumption. This assumption states that if the wife is indifferent between working or not working the husband should be indifferent as well (or, in above terms, there should be no discrete 'jump' in the labour supply function of the husband). Blundell *et al.* (2007) show that if there were a discrete 'jump' in the husband's preferences, there would be scope for a Pareto improving reallocation of the household's resources. This contradicts the assumption of Pareto efficiency.

Similarly, I denote the sharing rule in case of a non-working wife by

$$\rho(w_m, w_f, y) = \mathbf{K}'\mathbf{W}. \tag{15}$$

Letting $\mathbf{k}'\mathbf{W}$ represent the vector notation of the right-hand side of (9). Continuity on the participation frontier imposes the following restriction

$$\mathbf{K}'\mathbf{W} = \mathbf{k}'\mathbf{W} + r_m(\boldsymbol{\phi}'_f\mathbf{W}). \tag{16}$$

In (16) r_m is a parameter. The following relation between the parameters s_m and r_m from (14) and (16) is implied by the structure of the collective setting:

$$r_m = \frac{s_m b_4}{\Delta}. \tag{17}$$

In a similar way, I may introduce a switching regression for the wife if the husband does not work. Let the wife's labour supply function in this case be presented by

$$h_f(w_m, w_f, y) = \mathbf{\Phi}'_f\mathbf{W}, \tag{18}$$

and let the sharing rule be

$$\rho(w_m, w_f, y) = \tilde{\mathbf{K}}'\mathbf{W}. \tag{19}$$

The equivalents of (14) and (16) become

$$\boldsymbol{\Phi}_f'\mathbf{W} = \boldsymbol{\phi}_f'\mathbf{W} + s_f(\boldsymbol{\phi}_m'\mathbf{W}), \tag{20}$$

and

$$\tilde{\mathbf{K}}'\mathbf{W} = \mathbf{k}'\mathbf{W} + r_f(\boldsymbol{\phi}_f'\mathbf{Z}), \tag{21}$$

and the collective framework implies the following restriction between the parameters r_f and s_f:

$$r_f = \frac{s_f a_4}{\Delta}. \tag{22}$$

2.2. *Heterogeneity and Coherence*

2.2.1. *Introducing heterogeneity*

So far I have abstained from introducing observed and unobserved heterogeneity. The identification results by Donni (2003) are derived in the absence of heterogeneity. The theoretical framework by Blundell *et al.* (2007) does not incorporate heterogeneity but, in their empirical model, they include additive observed and unobserved heterogeneity in the labour supply functions and the sharing rule. The additive form of the heterogeneity ensures that the derived results for the identification of the sharing rule remain valid, as the additive constant of the sharing rule is not identified. It can easily be checked that a labour supply function that is additive in observed and unobserved heterogeneity is obtained if the preference parameter ϑ_j in (8) is specified as a linear combination of observed and unobserved heterogeneity. Alternatively, additive heterogeneity may come from additive heterogeneity in the sharing rule. Therefore, it is not possible to identify whether the additive heterogeneity in the labour supply function comes from heterogeneity in preferences for leisure or from heterogeneity in the sharing rule.

The available data form a panel. With subscript i and t I denote household i and year t, respectively. Since the panel is unbalanced, the number of time periods varies with i and I denote T_i. The number of households is indicated by N.

Let $\mathbf{z}_{itm}$ and $\mathbf{z}_{itf}$ be vectors of observed characteristics affecting male and female preferences, respectively. Let $v_{itj}, j = m, f$ denote unobserved heterogeneity, varying across households and years, and $\theta_{ij}, j = m, f$ denote unobserved time invariant heterogeneity for household member j. I extend (10) by writing

$$h_{itj} = \boldsymbol{\phi}_j'\mathbf{W}_{it} + \boldsymbol{\zeta}_j'\mathbf{z}_{itj} + \theta_{ij} + v_{itj}, j = m, f, \tag{23}$$

for $i = 1,\ldots, N$, $t = 1,\ldots, T_i$.

The analysis so far assumed the observability of wages, even if someone is not working. For the empirical analysis, I specify the following wage equation for $i = 1, \ldots, N$, $t = 1,\ldots, T_i$:

$$\ln w_{itj} = \boldsymbol{\eta}_j'\mathbf{x}_{itj} + \omega_{ij} + u_{itj}, j = m, f, \tag{24}$$

with x_{itj} and u_{itj} observed and unobserved characteristics, and ω_{ij} an unobserved random effect. Combining (23) with (14), (20) and (24) and denoting $d_{itj} = 1$ if household member j participates, $d_{itj} = 0$ otherwise, I may summarise the conditions for participation, conditional on the wages, for households $i = 1, \ldots, N$ and time periods $t = 1,\ldots,T_i$ as[11]

$$\begin{aligned} d_{itj} &= \iota(h^*_{itj} > 0), j = m,f, \\ h^*_{itf} &= \boldsymbol{\phi}'_f \mathbf{W}_{it} + \boldsymbol{\zeta}'_f \mathbf{z}_{itj} + \theta_{if} + v_{itf} + (1 - d_{itm}) s_f(\boldsymbol{\phi}'_m \mathbf{W}_{it} + \boldsymbol{\zeta}'_m \mathbf{z}_{itm} + \theta_{im} + v_{itm}), \\ h^*_{itm} &= \boldsymbol{\phi}'_m \mathbf{W}_{it} + \boldsymbol{\zeta}'_m \mathbf{z}_{itm} + \theta_{im} + v_{itm} + (1 - d_{itf}) s_m(\boldsymbol{\phi}'_f \mathbf{W}_{it} + \boldsymbol{\zeta}'_f \mathbf{z}_{itf} + \theta_{if} + v_{itf}). \end{aligned} \tag{25}$$

The equations for male and female working hours become:

$$\begin{aligned} h_{itj} &= h^*_{itj}, \quad \text{if} \quad h^*_{itj} > 0, \ j = m,f, \\ h_{itj} &= 0 \qquad \text{otherwise.} \end{aligned} \tag{26}$$

2.2.2. *Coherence of the model*

In the estimation of the model a complication arises that so far has not been addressed in any discussion of the collective model. The system (25) is not necessarily coherent: without any further restrictions, for given values of $(\mathbf{W}_{it}, \mathbf{z}_{itm}, \mathbf{z}_{itf}, \theta_{im}, \theta_{if}, v_{itm}, v_{itf})$ the model in (25) may generate multiple outcomes for the participation of husband and wife in a household. For instance, the conditions for $(d_{itm} = 0,\ d_{itf} = 1)$ and for $(d_{itm} = 1,\ d_{itf} = 0)$ in (25) may not be exclusive. This arises because in the collective model with non-participation, the labour market participation status of the other partner enters the equation for labour market participation. This problem has not been addressed by Donni (2003) or Blundell *et al.* (2007).[12] A statistical consequence of incoherence is that the probabilities of the four combinations of male and female participation may add up to an amount larger than 1. Estimation of a model that is incoherent in parts of the parameter space can lead to inconsistent estimates. In general, imposing coherence in a model like this is either quite complicated – see, for instance, Kooreman (1994) – or greatly reduces the generality of the model; see Heckman (1978). For the system (25), however, it can be shown that, apart from the trivial case $s_m s_f = 0$, coherence holds for more combinations of s_m and s_f. In the Appendix I show that the model is coherent for all combinations of s_m and s_f for which $|s_m s_f| < 1$.

2.2.3. *Estimation by (simulated) maximum likelihood*

Now the complete model is defined by (24)–(26). I estimate the model parameters by maximum likelihood.

For the time varying error terms $(v_{itm}, v_{itf}, u_{itm}, u_{itf})'$ I assume that they are independently and identically distributed across households and across time. Moreover, I assume them to be uncorrelated with the random effects $(\theta_{im}, \theta_{if}, \omega_{im}, \omega_{if})'$ and uncorrelated with the observed characteristics. The random effects are independently distributed across households and uncorrelated with the observed characteristics.

Throughout I assume that the errors of the wage equations and the labour supply equations follow a joint normal distribution:

[11] The equations (25) adjust the specifications (14) and (20) include additive observed and unobserved heterogeneity.

[12] In Blundell *et al.* (2007) the problem is implicitly circumvented as their model describes working hours and participation for the wife, but only participation of the husband. Therefore the labour market state of the wife does not enter the participation equation of the husband.

$$\begin{pmatrix} v_m \\ v_f \\ u_m \\ u_f \end{pmatrix} \sim \mathrm{N}\left(\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_m^2 & \sigma_{mf} & \sigma_{mu} & 0 \\ \sigma_{mf} & \sigma_f^2 & 0 & \sigma_{fu} \\ \sigma_{mu} & 0 & \tau_m^2 & \tau_{mf} \\ 0 & \sigma_{fu} & \tau_{mf} & \tau_f^2 \end{pmatrix}\right). \tag{27}$$

Note that I allow for non-zero correlation between the errors of the labour supply functions (parameter σ_{mf}). Thus I allow for common, household specific, unobserved characteristics in, say, preferences. Furthermore, I allow for correlation between the male (female) labour supply equation and the male (female) wage rates to correct for possible selectivity bias for the fact that wages are only observed for participants. Note that I restricted the correlation between the errors of the husband's (wife's) labour supply equation and the wife's (husband's) wage equation to zero.[13] Finally, I allow for non-zero correlation between the errors of the wages of husband and wife. This may capture common unobserved components in the wages of both partners, like for instance the effect of region, or a matching effect. I assume that the random effects are normally distributed:

$$\begin{pmatrix} \theta_{im} \\ \theta_{if} \\ \omega_{im} \\ \omega_{if} \end{pmatrix} \sim \mathrm{N}\left(\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_{\theta,m}^2 & \sigma_{\theta,mf} & \sigma_{\theta\omega,m} & \sigma_{\theta\omega,mf} \\ \sigma_{\theta,mf} & \sigma_{\theta,f}^2 & \sigma_{\theta\omega,fm} & \sigma_{\theta\omega,f} \\ \sigma_{\theta\omega,mu} & \sigma_{\theta\omega,fm} & \sigma_{\omega,m}^2 & \sigma_{\omega,mf} \\ \sigma_{\theta\omega,mf} & \sigma_{\theta\omega,f} & \sigma_{\omega,mf} & \sigma_{\omega,f}^2 \end{pmatrix}\right). \tag{28}$$

The construction of the likelihood function consists of the following steps. From (27), (28) and (26) the likelihood contributions for working hours, conditional on the wages w_{itm} and w_{itf} and random effects $(\theta_{im}, \theta_{if}, \omega_{im}, \omega_{if})'$ can be constructed. Next, the density function of wages, conditional on random effects, can be formed. To complete the likelihood contribution I integrate over random effects. For observations without observed values for the wages, which includes the non-participating men and women, I integrate over wages to complete the likelihood contribution. In the application I employ the smooth simulated maximum likelihood method (Börsch-Supan and Hajivassiliou, 1993) with 60 replications to integrate over wages and random effects. The coherence constraint $|s_m s_f| < 1$ will be imposed in the estimation of the model.

The model formulated in (26) and (25) may be estimated with information on both participation and working hours of husband and wife. In this respect, the model expands on the work by Blundell *et al.* (2007) and Donni (2003): both approaches suggest modelling collective labour supply by a switching regression model, using information on both working hours and participation of one partner and only on participation of the other partner. Thus, variation in working hours by the other partner is ignored.

2.2.4. *Sharing rule parameters that vary with marital status: estimation strategy*

In Section 2.1 I discussed the idea that the parameters of the sharing rule (9) may differ by marital status, and I may write $\kappa_j = \kappa_j(b)$, $j = 1,\ldots,7$, distinguishing married couples

[13] Note that this restriction is not necessary for identification but I did not find any *a priori* reason for allowing this correlation coefficient to vary. Also note that this restriction does not imply that, say, the distribution of v_m, conditional on (v_f, u_m, u_f) does not depend on u_f, since I allow for correlation between v_m and v_f and for correlation between v_f and u_f.

and unmarried couples. If preferences do not vary with marital status,[14] the parameters $\alpha_l^j, j = m, f$ do not depend on marital status. The parameters of the reduced form labour supply function, (10), will depend on marital status. Now that I have introduced heterogeneity, I can say more about the implementation of an estimation procedure that allows the parameters to vary with marital status. The additive heterogeneity $\zeta_j' z_{itj} + \theta_{ij} + v_{itj}$ can be attributed to both heterogeneity in preferences and heterogeneity in the level of the sharing rule κ_1: it cannot be identified, because the intercept of the sharing rule is not identified separately from preferences. However, if the parameters of the sharing rule are allowed to depend on marital status, in general the parameters of the wage rates and non-labour income, ϕ_j, $j = m, f$, vary not only with marital status but also with the parameters that affect heterogeneity. Not allowing for this flexibility may lead to spurious effects in the estimation of the parameters ϕ_j, $j = m, f$ and hence inconsistent estimates of the structural parameters $\alpha_l^j, j = m, f, l = 2, 3, 4$ and $\kappa_l(b)$, $l = 2, \ldots, 7$. To allow for this flexibility, I have adopted the following approach. I stratify the sample and estimate the model separately for married and unmarried couples. Next, I use a minimum distance estimator to estimate the parameters of the structural labour supply function and the parameters of the sharing rule. Also, I can use the minimum distance criterion to test the hypothesis that the structural labour supply parameters do not depend on marital status. Moreover, with the same parameter estimates of the full system (25), I can obtain different minimum distance estimates for the parameters κ_l, $l = 2, \ldots, 7$: estimates that do and estimates that do not vary with marital status. Thus, I can test the hypothesis of constant sharing rule parameters.

3. The Data

I use data from the Socio-Economic Panel (SEP). The SEP is a household survey collected by Statistics Netherlands. I use data for the years 1990 to 2001. In this period, households were interviewed on a yearly basis, every May. An obvious advantage of this household survey is that not only is the head of the household interviewed but detailed information is obtained about the partner as well. I made several selections to arrive at the dataset that I use in my analysis. For each year, I selected couples living together (either married or unmarried) without children, with the male in the age range of 22 to 60 and the female no older than 60. I excluded households in which either the husband or the wife was self-employed. Furthermore, I require the availability of information on the labour market state of both household members, the non-labour income and information on the level of schooling and the sector of education. I use information on hourly wage rates and working hours of both partners but I also use information on individuals with missing working hours and wage rates.[15] The pooled dataset contains 8,049 observations ('couple-years').

Table 1 contains descriptive statistics for the data. It shows that 84.4% of the male respondents are employed as are 70.3% of their female partners. In interpreting these

[14] The discussion in Section 2.1 showed that preferences may depend on marital status as the individual utility level is a determinant of the (net) gains of marriage. Thus, married individuals may have selected themselves into this state because of their preferences.

[15] In the estimation, likelihood contributions will be adjusted accordingly.

Table 1
Descriptive Statistics

Variable	Pooled data 8,049 Observations		Married couples 5,558 Observations		Unmarried couples 2,491 Observations	
	Husband	Wife	Husband	Wife	Husband	Wife
Employment status						
Employed	84.4%	70.3%	80.4%	62.0%	93.3%	88.7%
Not Employed	15.6%	29.7%	19.6%	38.0%	6.7%	11.3%
Education level						
Primary	7.3%	11.2%	9.2%	14.3%	3.0%	4.3%
Lower vocational	16.0%	23.4%	16.0%	26.7%	16.2%	16.0%
Intermediate	49.3%	42.5%	49.9%	40.7%	48.1%	46.6%
Higher Vocational	20.0%	18.2%	19.8%	15.1%	20.3%	25.3%
University degree	7.0%	4.4%	4.8%	3.1%	11.9%	7.4%
Education sector						
Technical	34.4%	5.3%	34.1%	4.5%	35.1%	7.1%
Economic/administrative	25.9%	24.5%	24.8%	23.8%	28.5%	26.1%
General (not specialised)	18.1%	30.2%	19.0%	32.7%	16.1%	24.7%
Services	21.5%	40.0%	22.1%	39.0%	20.2%	42.1%
Weekly working hours						
No. of observations	6,618	5,408	4,350	3,300	2,268	2,108
Mean	39.4	30.9	39.3	28.9	39.6	34.2
(Standard deviation)	(7.9)	(10.8)	(8.0)	(11.3)	(7.8)	(9.2)
Hourly wage rates						
No. of observations	6,258	4,997	4,113	3,009	2,145	1,988
Mean (Guilders)	21.8	18.2	23.0	18.7	19.1	17.4
(Standard deviation)	(7.7)	(6.4)	(8.0)	(6.7)	(6.5)	(6.0)
Age						
Mean	40.8	38.7	45.1	43.1	31.2	28.8
(Standard deviation)	(12.4)	(12.5)	(11.6)	(11.7)	(7.7)	(7.9)
Household level variables						
Non-labour income						
Household level, weekly						
Mean (guilders)	37.7		36.8		39.8	
Standard deviation	(94.8)		(93.2)		(98.2)	
Employment status						
Both partners working	64.3%		55.4%		84.0%	
Husband working, wife not	20.2%		25.1%		9.3%	
Wife working, husband not	6.0%		6.6%		4.7%	
Both not working	9.5%		12.9%		2.0%	
Marital status						
Married	69.1%		100%		0%	

numbers please recall that couples without children were selected. Therefore, the percentages of males and females working are relatively high in this sample. 64.3% of the households are dual earner couples. In 20.2% of the households the husband is the sole breadwinner. For 9.5% of the households none of the members is working, whereas in 6.0% of the households only the wife works.

The males in the sample are on average more highly educated than the females. I also have information about educational criteria and there are some typical differences between males and females. There are few women with a technical type of education whereas the majority of the men followed a technical education. Most women are educated for the service sector. There are also more women without specialisation in education. The mean age for males is about 2 years higher than for females, which is quite common for married couples.

The mean weekly number of working hours is 39 for males and 31 for females. The males have an hourly wage rate that on average is almost 4 guilders higher than the wage rate of females. The non-labour income includes interest income, income out of real estate, rent subsidy, income out of life insurance ('lijfrente'), gifts by family, income out of profits and scholarships. In the survey it is measured on a yearly basis and in Table 1 it is converted to guilders per week. The average is about 34 guilders a week, and there is quite some variation in it, with some households reporting much higher amounts, and some households reporting not to have received any non-labour income.

A comparison of the sub sample of the married and the unmarried shows that the unmarried are on average younger, there are more dual earner couples among the unmarried, education levels are higher among the unmarried and education differences between men and women are on average smaller for unmarried couples.

The pooled data contain observations on 5,558 married couples and 2,491 unmarried couples. The sub sample of married couples contains 1,514 different households and the sub sample of unmarried couples consists of 875 different households. Households appear at least once and at most 12 times in the sample. About 70% of the households appear more than once. For various reasons only a limited number of the households appear in all of the 12 years. These reasons include attrition (households either leave the panel or do not satisfy my selection criteria anymore), item non-response (relevant information is missing), wave non-response, or households may be new entrants in the panel. Throughout I maintain the assumption that non-response is independent of hours, wage rates and employment status (as it is defined in this analysis). Note that attrition of households may also occur if a couple splits up: a household is considered the same through different waves if both of the partners remain the same two persons.

4. Results

I have estimated the model defined in (24)–(28) by the method of simulated maximum likelihood (SML). I use 60 replications to simulate wages (for individuals without an observed wage) and the random effects. To allow the underlying parameters of the sharing rule to differ by marital status, I estimate the model separately for unmarried and married couples. Note that additive heterogeneity in the (reduced form) labour supply function can come both from (observed and unobserved) heterogeneity in preferences and from additive heterogeneity in the sharing rule. Therefore, unless one is willing to make arbitrary exclusion restrictions about the source of heterogeneity, a sharing rule that differs by marital status requires that all parameters of the labour supply functions are different for the married and the unmarried.[16]

I use a minimum distance estimator to impose and test equality of the parameters of the structural labour supply function (6) by marital status. I also use a minimum distance estimator to impose and test equality of the parameters of the sharing rule (9) by marital status.

[16] This way, I aim to avoid spurious effects: if only the parameters of the wage rates and non-labour income would differ by marital status, these parameters may just react to neglected flexibility in heterogeneity.

4.1. *Parameter Estimates of the 'Reduced Form' Labour Supply Function*

Table 2 presents the estimates of the parameters of the labour supply function of husband and wife by marital status. These are the estimates of the parameters a_j and b_j, $j = 1,\ldots,7$ in (10).[17] The parameters of the remaining coefficients are presented in the Appendix. Table A1 shows the coefficients of the observed characteristics included in the labour supply function, Table A2 shows the coefficients of the wage equations, and Table A3 presents the coefficients of the covariance matrices. I use the information on education sectors as an exclusion restriction in the wage equations. Thus, I assume that the individual preferences for leisure do not vary by sector. However, wages are likely to differ by sector. The service sector is the reference category. In the wage equation, the highest level of education is the reference level. In the labour supply functions I have merged the two higher education levels to one category, since the highest education group is small. I have also included time dummies in the wage equations.

To evaluate the full effect of a change in the husband's wage rate on his supply of labour I computed the marginal effect $a_2 + a_4 w_f/(w_m + w_f)^2 + 2a_5 w_m$ for all couples with observed wage rates in the sample (separately for the married and the unmarried). For unmarried men, this effect is negative for 7.4% of the cases, while for married men, the percentage with a negative wage effect is 67.7%. The latter indicates a backward bending labour supply curve for married men.[18] The marginal effect of the wife's wage rate on the husband's labour supply is determined by $a_3 - a_4 w_m/(w_m + w_f)^2 + 2a_6 w_f$. For the married, this is positive for 98% of the couples, suggesting that married men tend to increase working hours upon a wage increase of the wife. For the unmarried, the percentage with a positive effect of the wife's wage rate on the husband's labour supply is 79%. Non-labour income has a significant negative effect on the husband's supply of labour, both for married and unmarried men.

The marginal effect of a change in the wife's wage rate on her labour supply is $b_3 - b_4 w_m/(w_m + w_f)^2 + 2b_5 w_f$. Both for married and unmarried women, this is positive at mean wage rates but for high wage rates the negative backward bending effect dominates. Counting frequencies, the backward bending effect dominates for 2.2% of the married women and for 0.8% of the unmarried. Both for married and unmarried women I find that the marginal effect of the husband's wage rate on their labour supply is negative for about 98% of the wage rates in the sample. For married women non-labour income has a negative effect on the wife's labour supply for unmarried but a positive effect for the married.

The estimates of a_4 and b_4, the coefficients of the wage rate fraction, have the opposite sign, as required by the underlying collective model.

The parameter estimates of s_m and s_f, that determine the husband's (wife's) labour supply if the wife (husband) does not work, are positive and negative, respectively. In the estimated values, the coherence constraint $|s_m s_f| < 1$ is satisfied. The parameter s_m is significant and, moreover, its impact is much larger than s_f, which is not estimated

[17] Note that due to normalisations, Table 2 presents the coefficients $1000a_7$,$1000b_7$,$10a_5$,$10a_6$,$10b_5$, and $10b_6$.

[18] With different data from a different time period, Kapteyn *et al.* (1990) also find evidence of a backward bending labour supply curve for men, whereas backward bending is absent for women.

Table 2

Simulated Maximum Likelihood Estimates: Labour Supply Equations

	Unmarried		Married	
Variable	Men	Women	Men	Women
Intercept	−11.829	31.319**	16.468**	30.374**
	(9.562)	(11.121)	(8.271)	(6.411)
Wage husband	0.291	0.342	0.192	0.192
	(0.394)	(0.543)	(0.274)	(0.241)
Wage wife	1.278**	0.102	0.422	0.900**
	(0.457)	(0.543)	(0.305)	(0.270)
$w_m/(w_m + w_f)$	50.619**	−38.265**	8.425	−39.761**
	(17.408)	(22.468)	(14.168)	(11.400)
Wage husband squared/10	−0.120**	−0.016	−0.085**	0.004
	(0.043)	(0.061)	(0.027)	(0.024)
Wage wife squared/10	−0.132**	−0.064	−0.054	−0.128**
	(0.063)	(0.067)	(0.033)	(0.034)
Non-labour income/1000	−9.728**	−7.157**	−4.709**	5.902**
	(2.140)	(2.304)	(2.165)	(1.407)
$s_j (j = m,f)$	118.095**	−0.005	40.950**	−0.024
	(42.895)	(0.069)	(3.556)	(0.075)

** (*): significant at 5% (10%) level. Standard errors in parentheses.

significantly. I comment on these differences when I discuss the results of the sharing rule.[19]

4.2. *Estimates of Structural Labour Supply and the Sharing Rule*

Using the estimates of the parameters of the reduced form labour supply equations from Table 2, the parameters of the individual labour supply functions (6) and the sharing rule (9) can be computed using the expressions in (11) and (12). Thus, the total effects of the wage rates and non-labour income on labour supply can be decomposed into effects arising from individual preferences and into the effects arising from the sharing rule.[20] This decomposition can be done separately for the married and unmarried. First I assume that the parameters of the wage rates and non-labour income of the structural labour supply equations, (6), are the same for the married and the unmarried[21] but the sharing rule parameters differ by marital status. I impose this restriction and obtain parameter estimates using a minimum distance estimator. Next, I use a second minimum distance estimator to obtain estimates under the additional assumption that the parameters k_2 to k_7 of the sharing rule are the same for the married and the unmarried. Imposing restrictions may lead to more precise estimates but only

[19] For reasons of comparison, I also obtained estimates of the reduced form parameters ignoring non-participation: I re-estimated the model using dual earner couples only. The key results are in Appendix Table A4, which is the equivalent of Table 2. I find different coefficient estimates for the parameters of the wage rates and non-labour income. For instance, the coefficients of non-labour income in Table A4 have the same sign as in Table 2 but the estimates seem to be biased towards zero.

[20] Recall that this decomposition is based on the assumption of egoistic preferences, while I have assumed that the wage fraction enters the sharing rule but not the structural labour supply function.

[21] To be more precise, I assume that the parameters $\alpha_l^j, l = 2, 3, 4, j = m, f$ do not differ by marital status. I do not impose any restrictions on the intercept and the parameters of observed and unobserved heterogeneity.

Table 3

Parameters Structural Labour Supply by Marital Status

Variable	Directly computed from estimates in Table 2: Unmarried	Married	Minimum distance, same labour supply, different sharing rule All	Minimum distance, same labour supply, same sharing rule All
Men				
Non-lab. inc(α_2^m)	−0.019**	−0.0035	−0.0066**	−0.0072**
	(0.0056)	(0.0028)	(0.0014)	(0.0015)
(Own) Wage (α_3^m)	0.74	0.23	0.43**	0.26*
	(0.45)	(0.20)	(0.11)	(0.15)
(Own) wage squared (α_4^m)	−0.014**	−0.0084**	−0.010**	−0.0080**
	(0.068)	(0.0026)	(0.0015)	(0.0020)
Women				
Non-lab. inc(α_2^f)	−0.015**	−0.016	−0.015**	−0.0094**
	(0.0046)	(0.035)	(0.0041)	(0.0035)
(Own) Wage (α_3^f)	1.1	2.9	1.9**	1.9**
	(1.3)	(2.3)	(0.21)	(0.14)
(Own) wage squared (α_4^f)	−0.016	−0.038	−0.024**	−0.024**
	(0.23)	(0.14)	(0.0034)	(0.0021)

** (*): significant at 5% (10%) level. Standard errors in parentheses.

make sense if the restrictions imposed are valid. The value of the objective function of the minimum distance estimator follows the chi-squared distribution and can be used to test the restrictions.

Tables 3 and 4 show the parameters estimates of the individual labour supply functions (6) and the estimates of the sharing rule (9), respectively. They contain both the estimates that do not impose any restrictions between the married and the unmarried, and the estimates that impose restrictions on the preferences parameters and the sharing rule. The chi-squared statistic, obtained in the minimum distance estimation procedure is 56 if only the parameters of the wage rates and non-labour income in the labour supply equations are restricted to be equal for the married and the unmarried (6 restrictions). It takes the value 96 if the parameters of the sharing rule are restricted to be equal as well (14 restrictions in total). In both cases, the restrictions are rejected.[22] Therefore, I prefer the estimates obtained by separate estimation for the married and unmarried. A comparison with the restricted estimates may reveal the possibly distorting influence of the imposition of restrictions between the married and unmarried on key outcomes.

The first two columns of Table 3 contain the parameters of the individual labour supply functions (6) for the unmarried and married.[23] Not all the parameters are estimated precisely. However, regularity conditions are satisfied. Since the coefficient of non-labour income is negative (both for married and unmarried men and women) the estimates imply that leisure is a normal good.

[22] In Section 2.1 I argued that there may be two reasons why outcomes may differ by marital status. Marital status can influence the bargaining power within the household but the decision to marry is also influenced by the utility obtained while being single. The rejection of the restrictions shows that not only the sharing rule parameters differ by marital status, but also the preference parameters that enter the individual labour supply functions.

[23] Squared wages and non-labour income are, unlike in Table 2, not normalised in Tables 4 and 5.

Table 4

Parameters of the Sharing Rule by Marital Status (see Equation 9)

Variable	Directly computed from estimates in Table 2		Minimum distance, same labour supply, different sharing rule		Minimum distance, same labour supply, same sharing rule
	Unmarried	Married	Unmarried	Married	All
Men					
Wage husband, (k_2)	24	12	76*	26	36
	(32)	(34)	(41)	(18)	(24)
Wage wife, (k_3)	−66**	−122	−160**	−86	−130**
	(29)	(171)	(54)	(31)	(47)
$w_m/(w_m + w_f)$, (k_4)	−2,637**	−2,436	−5,578**	−3,308**	−5,088**
	(1,096)	(5,617)	(2,124)	(1,249)	(1,984)
Sqr. wage husband, (k_5)	−0.11	0.022	−0.56	−0.10	−0.087
	(0.4)	(0.14)	(0.42)	(0.17)	(0.22)
Sqr. wage wife, (k_6)	0.69**	1.6**	1.5**	0.91**	1.2**
	(0.38)	(0.2)	(0.6)	(0.39)	(0.5)
Non-labour income, (k_7)	0.51**	1.4*	0.51**	1.4**	1.3**
	(0.17)	(0.8)	(0.20)	(0.15)	(0.18)
r_m	−6,152	−11,841	−19,278**	−6,138**	−5,735**
	(5,507)	(15,558)	(7,460)	(1,368)	(1,259)
r_f	−0.34	−1.5	−0.16	−1.8	−0.18
	(4.7)	(5.7)	(4.7)	(5.1)	(5.3)

** (*): significant at 5% (10%) level. Standard errors in parentheses.

For unmarried men, the backward bending labour supply effect becomes dominant from wage rates with values from around the sample mean on, whereas for married men the backward bending effect is also dominant for wage rates below the sample mean. For women, the backward bending effect is much weaker and there is a positive effect of wage rates for almost all women in the sample.

Table 4 contains the estimates of the sharing rule (9). The estimates in the first two columns (for the married and the unmarried) have been directly computed from the reduced form parameters in Table 2. Thus, no restrictions are imposed between any of the parameters of the married and the unmarried. The third and fourth columns are obtained by the first minimum distance estimator, that restricts the parameters of the wage rates and non-labour income in the structural labour supply function to be equal for the married and the unmarried. However, the parameters of the sharing rule differ by marital status. The final column shows the results obtained with the second minimum distance estimator. Both the parameters of the structural labour supply function and the parameters of the sharing rule are restricted to be equal for the married and the unmarried. In general, coefficients have the same sign for the different methods of estimation, but the quantitative impact differs. For the unmarried, the coefficient of non-labour income is around 0.5, indicating that unmarried couples seem to share their non-labour income fifty-fifty. For the married, I find a coefficient slightly larger than, but not always significantly different from, 1. This suggests that in married couples a substantial part of the non-labour income is assigned to the husband.[24] This

[24] Note that this result is similar to an earlier version of the article, where married and unmarried were pooled in the estimation, and the coefficient of non-labour income was about 1 as well. The present coefficients reveal heterogeneity by marital status.

difference between the married and the unmarried is not revealed if the coefficients of the sharing rule are restricted to be equal: the final column of Table 4 shows a coefficient of non-labour income that is not significantly different from one.

The marginal effect of a change in w_m on the sharing rule is $k_2 + k_4 w_f/(w_m + w_f)^2 + 2k_5 w_m$. I computed these marginal effects for all couples with observed wage rates in the sample. I counted the percentage of households for which the effect is positive. Depending on which parameter estimates and subsample from Table 4 are considered, the percentage with a negative impact ranges from 96 to 98. Thus, the wife's share increases upon an increase in the husband's wage rate. The consequence of the increase in the wife's share is that her supply of labour will diminish since leisure is a normal good to her.

The marginal effect of the wife's wage rate on the sharing rule is $k_3 - k_4 w_m/(w_m + w_f)^2 + 2k_6 w_f$. This is negative for the majority of the observations. Depending on which parameter estimates and subsample from Table 4 are considered, the percentage with a negative impact ranges from 79 to 98. Thus, an increase in the wife's wage rate apparently leads to a decrease in the husband's share for most of the cases.

The negative impact of the wife's wage rate on the share of the husband is easily interpretable in terms of an increase in the wife's bargaining power as a result of the increase in her wage rate. The husband's resources decrease and he increases his supply of labour to compensate for that. The negative impact of the husband's wage rate on his share is beneficial to the wife. Here a household income effect dominates, allowing the wife to consume more leisure. The total effect of the increase of the husband's wage rate on his welfare level is ambiguous. The decrease in his share decreases his leisure and therefore decreases his utility level. On the other hand, the higher wage rate has, at given working hours, a positive impact on his private consumption, which increases his utility level. A possibly backward bending labour supply curve may increase his leisure, with a positive consequence for his utility level.

To evaluate the eventual impact on utility, I have computed the marginal effects of the wage rates and the non-labour income on the indirect utility function, substituting the appropriate share for non-labour income.[25] I computed the marginal effects for all the observations with observed wage rates and for 60 replications of the random effects.[26] Table 5 reports the sample percentage of men and women experiencing an increase in utility due to a change in either of the wage rates or the non-labour income. For almost all men in the sample, the utility level increases if the men's wage rate goes up, irrespective of the estimates looked at. Also for almost all women the utility level increases as a result of an increase in the husband's wage rate, which reflects the fact that the wife's share increases for almost all women.

An increase in the wife's wage rate leads to an increase in the husband's utility level for only a small fraction of the sample. This suggests that the wife's bargaining power

[25] The indirect utility function belonging to the specification (7) reads $v(w_j, \mu_j) = (\vartheta_j + \mu_j + \delta_j w_j + 1/2\, w_j^2 \gamma_j) \exp(\beta_j w_j), j = m, f$. The parameters can be computed from the estimates, as indicated below (8). Additive heterogeneity enters by ϑ_j.

[26] Thereby assuming that the random effects are part of preferences and/or the sharing rule.

Table 5

Sample Percentages of Individuals with an Increase in Utility

	Estimation method					
	Directly computed from estimates in Table 2		Minimum distance, same labour supply, different sharing rule		Minimum distance, same labour supply, same sharing rule	
	Percentage men with increase in utility					
Increase in	Unmarried	Married	Unmarried	Married	Unmarried	Married
Wage rate husband	99.4	100	100	100	99.9	100
Wage rate wife	20.9	2.4	5.4	15.1	15.1	11.3
Non-labour income	100	100	100	100	100	100
	Percentage women with increase in utility					
Increase in	Unmarried	Married	Unmarried	Married	Unmarried	Married
Wage rate husband	97.8	97.5	95.9	97.1	97.7	96.2
Wage rate wife	99.9	100	99.3	100	100	100
Non-labour income	100	0	100	0	0	0

Sample Percentages of Individuals with an Increase in Utility in Response to an Increase in Wage Rates and Non-labour Income.
Computations for couples with observed wages.

increases as her wage rate increases. The utility level increases for almost every woman in the sample.

The largest difference between the unmarried and the married is shown by an increase in the household's non-labour income. Both for the unmarried and the married, the husband's share is increased as a result of an increase in the non-labour income. Since leisure is a normal good, his supply of labour decreases, which has a positive direct effect on his utility. His total earnings will decrease due to the decrease in working hours. The total effect is an increase in utility for all unmarried and married men. For unmarried women, the share is also increased due to an increase in the household's non-labour income. I also find an increase in the utility of all unmarried women. For married women, the share decreases. It should be noted, however, that since the coefficient of the non-labour income is not always significantly different from one, the women's share may not be affected significantly. The utility of all married women decreases. This suggests that unmarried women have a stronger bargaining position than married women. Assigning the same sharing rule parameters to married and unmarried women leads to the wrong conclusion that the utility level of unmarried women decreases as well.

From estimates in Table 4 I also compute the sharing rule for the case in which the husband (wife) works and the wife (husband) not (see (16) and (21)). I have done this explicitly for the unrestricted estimates. The results are in Table 6. Notably the sharing rule for a working husband and a non-working wife is different from the rule in Table 4, which is due to the large value of the estimate of r_m in (16). The estimate of r_f in (21) is much smaller and is not significantly different from zero. Reservation wages of non-working males may be much lower and show

Table 6

The Sharing Rule Outside the Participation Frontier

	Husband works Wife not Equation (16)		Wife works Husband not Equation (21)	
Variable	Unmarried	Married	Unmarried	Married
Wage husband	−2,079	−2,267	23**	12
	(4,992)	(5,398)	(7)	(80)
Wage wife	−695	−10,777	−67**	−123
	(2,936)	(12,175)	(3)	(172)
$w_m/(w_m + w_f)$	232,772	468,391	2,654**	−2,449
	(337,906)	(728,362)	(1,100)	(5,638)
Sqr. wage husband	10	−4	−0.11	0.034
	(44)	(25)	(24)	(5)
Sqr. wage wife	40	153	0.69*	1.6
	(33)	(182)	(0.38)	(2.0)
Non-labour income	45	−69	0.51**	1.4*
	(41)	(91)	(0.17)	(0.8)

** (*): significant at 5% (10%) level. Standard errors in parentheses.
Estimates computed from Table 2.

less variation than reservation wages of non-working females. This may explain why the sharing rule for working males with non-working wives differs much more from the sharing rule in Table 4. I have determined that for almost all wage rates in the sample range an increase in the wage rate of the husband now leads to an increase in the 'share' of the husband. This may reflect a higher bargaining power of the husband if the wife is not working. Moreover, a non-working woman cannot reduce working hours if the husband transfers his wage increase to his wife. The latter may be a motivation for a husband with a working wife to increase her share upon an increase in her wage rate. The share of the husband decreases if the wage rate of the non-working wife increases, both for married and unmarried couples. Thus, the wage rate of a non-working wife may still function as a threat point. However, the relatively large standards error indicate that it is hard to obtain accurate estimates of the sharing rule for households with a non-working wife. This may be related to the small fraction of these households in the sample.

4.3. *Elasticities*

The qualitative implications of the estimation results obtained by different methods do not deviate much from each other, except for the specification with a common sharing rule for the married and the unmarried. However, the restrictions imposed in the minimum distance estimation have been rejected. Computation of the elasticities for the various estimates may reveal the quantitative implications of imposing these restrictions. Tables 7 and 8 report elasticities of labour supply with respect to the wage rates of both spouses and non-labour income. The elasticities are computed separately for the married and the unmarried, and evaluated in the sample means of the corresponding subsamples. Table 7 shows total effects of preferences and the sharing rule. The elasticities for the married and the unmarried in the first two columns are directly

Table 7

Elasticities: Total Effects of Wage Rates and Non-labour Income on Labour Supply

	Estimation method					
	Reduced form estimates from Table 2		Minimum distance, same labour supply, different sharing rule		Minimum distance, same labour supply, same sharing rule	
	Men: elasticities of labour supply					
Elasticity w.r.t.	Unmarried	Married	Unmarried	Married	Unmarried	Married
Wage rate husband	0.24**	−0.057*	0.071	0.028	0.092	0.036
	(0.04)	(0.032)	(0.093)	(0.058)	(0.086)	(0.090)
Wage rate wife	0.038	0.052**	0.078**	0.023	0.040	0.052
	(0.030)	(0.021)	(0.023)	(0.062)	(0.067)	(0.078)
Non-labour income	−0.0096**	−0.0042**	−0.0033**	−0.0082**	−0.0092**	−0.0083**
	(0.0021)	(0.0019)	(0.0017)	(0.0016)	(0.0014)	(0.0013)
	Women: elasticities of labour supply					
Elasticity w.r.t.	Unmarried	Married	Unmarried	Married	Unmarried	Married
Wage rate husband	−0.12**	−0.18**	−0.15**	−0.16**	−0.17**	−0.17**
	(0.04)	(0.03)	(0.04)	(0.03)	(0.03)	(0.03)
Wage rate wife	0.22**	0.61**	0.36	0.60**	0.48**	0.55**
	(0.07)	(0.04)	(0.31)	(0.19)	(0.14)	(0.16)
Non-labour income	−0.0081**	0.0071**	−0.0080**	0.0071**	0.0033	0.0035
	(0.0023)	(0.0013)	(0.0044)	(0.0032)	(0.0026)	(0.0023)

** (*): significant at 5% (10%) level. Standard errors in parentheses.

based on the reduced form estimates in Table 2. To compute the elasticities of labour supply for the minimum distance estimates, I have inserted the sharing rule estimates from Table 4 in the structural labour supply function, based on the corresponding estimates in Table 3.

None of the estimation methods shows much sensitivity of male labour supply to either the husband's wage rate or the wife's wage rate. The unrestricted estimates show that married men are less sensitive with respect to non-labour income than unmarried men. I do not find this when restrictions are imposed. For married women, their wage elasticities, both with respect to their own wage rate and their husbands', are quite stable across estimation methods. The wage elasticity of the unmarried women's own wage rate is affected most by imposing restrictions by marital status. The largest deviation due to imposing restrictions is found for the elasticities of female labour supply with respect to non-labour income. Overall, the labour supply of the wife is more sensitive with respect to changes in the wage rates than the labour supply of the husband. The order of magnitude of the wage elasticities for the women is comparable to the values usually found in the empirical literature.[27] Women are more responsive to changes in the partner's wage rate than men.

[27] For instance, Bloemen and Kapteyn (2008) estimate labour supply elasticities for married women with a Dutch data set, using a wide variety of estimation methods.

Table 8

Elasticities of the Structural Labour Supply Function

	Estimation method					
	Reduced form estimates from Table 2		Minimum distance, same preferences different sharing rule		Minimum distance, same preferences same sharing rule	
	Men: elasticities of labour supply					
Elasticity w.r.t.	Unmarried	Married	Unmarried	Married	Unmarried	Married
Wage rate	0.098	−0.085*	0.013	−0.026	−0.021	−0.058*
husband	(0.099)	(0.050)	(0.026)	(0.027)	(0.035)	(0.035)
Non-labour	−0.019**	−0.0031	−0.0065**	−0.0059**	−0.0071**	−0.0064**
income	(0.006)	(0.0025)	(0.0014)	(0.0012)	(0.0014)	(0.0013)
	Women: elasticities of labour supply					
Elasticity w.r.t.	Unmarried	Married	Unmarried	Married	Unmarried	Married
Wage rate	0.25	0.95	0.56**	0.67**	0.54**	0.65**
wife	(0.73)	(0.87)	(0.02)	(0.06)	(0.004)	(0.05)
Non-labour	−0.017**	−0.020	−0.017**	−0.018**	−0.011**	−0.011**
income	(0.005)	(0.043)	(0.005)	(0.003)	(0.004)	(0.004)

** (*): significant at 5% (10%) level. Standard errors in parentheses.

Table 8 shows the elasticities of the structural labour supply functions. These are conditional elasticities, since they are conditional on the 'share'. As the 'share' itself is not observed and identified only up to an additive constant, I used the household's non-labour income as a base value to compute the elasticity of labour supply with respect to the 'share'. The elasticities show the sensitivity of labour supply that can be assigned to preferences. The difference in the elasticities between Tables 7 and 8 is the effect that is due to the sharing rule. The estimates of the structural labour supply function obtained with the minimum distance estimator are the same for the unmarried and the married. The difference in the elasticities comes from differences in sample means for the married and the unmarried. The elasticities of labour supply of men with respect to their wage rate again reveal the insensitivity of male labour supply with respect to the wage rate. For female labour supply, the wage elasticities are not precisely estimated if no restrictions are imposed. The elasticities of female labour supply with respect to non-labour income does not differ much across estimation methods.

5. Conclusions

I have specified an empirical model of collective household labour supply that allows for non-participation of husband and wife. In my model I use information on both the participation and the working hours of husband and wife. To make this possible, particular attention needs to be paid to the coherence of the model. For my specification I derive a restriction on the parameter space such that coherence of the model is satisfied.

As a basis for my model I specify the labour supply function, based on individual preferences, as a function that is quadratic in wage rates and linear in non-labour income. The specification is flexible with in wage rates and allows for a backward bending labour supply function.

I estimate the model with data from the Dutch Socio Economic Panel (SEP) over the period 1990–2002. I allow the parameters of the sharing rule to be different for married and unmarried couples. With a minimum distance estimator I impose and test restrictions between the parameters of the married and the unmarried.

The estimates show that labour supply is backward bending for married men and to a lesser extent also for unmarried men. For women, I do not find a backward bending labour supply curve. The estimates satisfy some basic regularity conditions: leisure is a normal good for men and women, irrespective of their marital status, and a cross equation condition on the parameters that is implied by the collective model is satisfied.

The share of the husband decreases if the wage rate of the husband increases. This allows the wife to consume more leisure. For married men, the backward bending effect on labour supply is stronger, so the husband of a married couple may increase his leisure as well. For unmarried men, the labour supply is more likely to increase. I have shown that the utility level of both husband and wife increases as a result of the increase in the husband's wage rate. Apparently the increase in the husband's wage rate causes an increase in total household income and makes everyone in the household better off.

The share of the husband (wife) decreases (increases) if the wage rate of the wife increases. The husband will supply more labour. For women, the net result is also an increase in the supply of labour. The utility of the husband decreases and the utility of the wife increases due to an increase in the wife's wage rate. This is consistent with the interpretation that the wife's threat point increases due to an increase in her wage rate, and an improvement of her bargaining position. These qualitative effects of changes in wage rates on the sharing rule are the same for the married and the unmarried.

The effect of non-labour income differs by marital status. An increase in non-labour income is split between husband and wife in unmarried couples, whereas it is assigned to the husband in married couples. This suggests that unmarried women have a better bargaining position than married women. The results show that wrong conclusions are drawn for unmarried women if the sharing rule parameters are restricted to be the same for married and unmarried women. These restrictions are also formally rejected by the minimum distance criterion.

I have restricted the parameters of the wage rates and non-labour income in the structural labour supply function to be equal for the married and the unmarried. This leads to more precise estimates of the parameters. But these restrictions are formally rejected as well.

I can derive the parameters of the sharing rule of households with one non-employed member. The sharing rule of a working wife and a non-working husband is not significantly different from the sharing rule of a dual earner couple. The reason for this may be that reservation wages of men tend to be very low and show little variation. The sharing rule for a working husband with a non-working wife differs considerably from the sharing rule for working partners. An increase in the husband's wage rate now leads to an increase in the husband's 'share'. This suggests an increase in bargaining power of the husband if the wife is not working. An increase in the wife's wage rate still leads to a decrease in the husband's share.

Computation of elasticities shows that male labour supply is less sensitive to changes in wage rates than female labour supply. Imposing restrictions between the parameters of the married and the unmarried may lead to underestimation of the wage elasticity of unmarried men. The wage elasticities of married women are quite stable, irrespective of whether or not restrictions between the married and the unmarried are imposed.

Some comments are in order. Although the reduced form labour supply functions can be consistent with more general preferences, the decomposition into the structural labour supply equations and the sharing rule is based on the assumptions of egoistic preferences and the absence of public goods. Here I followed the theoretical results by Chiappori (1988) and Donni (2003). But egoistic preferences rule out that leisure time of spouses can be complements, or that non-market time of spouses can be substitutes. The model does not incorporate fixed costs of work, or the influence of demand side conditions on employment. The latter implies that no distinction is made between non-participation and involuntary unemployment. Incorporating fixed costs will make it harder to develop a statistical model that is continuous around the participation frontier. In estimating the sharing rule, I use non-labour income at the household level. Thus, I have not used information on ownership of underlying income sources to identify the shares assigned to each spouse. Models of labour supply and marriage (or matching) suggest the existence of public or shared goods. Becker (1973) argues that economies to scale in household production may lead to specialisation in time allocation between spouses. My empirical analysis allows for different outcomes by marital status but I do not model couple formation itself.[28] These considerations suggest that there is room for extensions of the basic model in several directions. In empirical work, identification of alternative models is a key issue, which sometimes requires the imposition of alternative assumptions or restrictions, but, in the better case, may be achieved by collecting the appropriate information in household surveys.

Appendix 1. On the Coherence of the Model

In this Appendix I show how I may derive restrictions on the parameters s_m and s_f in order to get coherence. To simplify notation, I denote the right hand side of (23) by $\mathbf{X}_j'\boldsymbol{\beta}_j$ for the purpose of this exercise:

$$\mathbf{X}_j'\boldsymbol{\beta}_j = \boldsymbol{\phi}_j'\mathbf{W} + \boldsymbol{\zeta}_j'\mathbf{z} + v_j,\ j = m, f. \tag{29}$$

Thus, I may rewrite model (25) as

$$\begin{aligned} d_m^* &= \mathbf{X}'_m\boldsymbol{\beta}_m + (1 - d_f)s_m\mathbf{X}'_f\boldsymbol{\beta}_f, \\ d_f^* &= \mathbf{X}'_f\boldsymbol{\beta}_f + (1 - d_m)s_f\mathbf{X}'_m\beta_m, \\ d_j &= \imath(d_j^* > 0). \end{aligned} \tag{30}$$

First, let me explain what coherence implies in the context of this model. Consider the case in which both spouses choose to work: $d_m = 1,\ d_f = 1$. The model in (30) generates this outcome if

[28] See e.g. Manser and Brown (1980).

both $\mathbf{X}'_m\boldsymbol{\beta}_m > 0$ and $\mathbf{X}'_f\boldsymbol{\beta}_f > 0$. The model would be incoherent if the model can generate a second outcome as well, for instance, $d_m = 0$, $d_f = 0$ by $\mathbf{X}'_m\boldsymbol{\beta}_m + s_m\mathbf{X}'_f\boldsymbol{\beta}_f \leq 0$ and $\mathbf{X}'_f\boldsymbol{\beta}_f + s_f\mathbf{X}'_m\boldsymbol{\beta}_m \leq 0$. Now, concentrate on this possible source of coherence, since the other outcomes, $d_m = 1$, $d_f = 0$ and $d_m = 0$, $d_f = 1$ cannot be generated by the model simultaneously with $d_m = 1$, $d_f = 1$. Consider when the events

$$\begin{aligned} \mathbf{X}'_m\boldsymbol{\beta}_m &> 0,\\ \mathbf{X}'_f\boldsymbol{\beta}_f &> 0, \end{aligned} \tag{31}$$

and

$$\begin{aligned} \mathbf{X}'_m\boldsymbol{\beta}_m + s_m X'_f\boldsymbol{\beta}_f &\leq 0,\\ \mathbf{X}'_f\boldsymbol{\beta}_f + s_f\mathbf{X}'_m\boldsymbol{\beta}_m &\leq 0, \end{aligned} \tag{32}$$

may occur simultaneously. Note that (31) and (32) cannot occur together if either $s_m \geq 0$ or $s_f \geq 0$.[29] The two events could occur, though, if both $s_m < 0$ and $s_f < 0$. Combining (31) and (32) yields two conditions:

$$s_m < -\frac{X'_m\beta_m}{X'_f\beta_f}(< 0), \tag{33}$$

and

$$-\frac{\mathbf{X}'_m\boldsymbol{\beta}_m}{\mathbf{X}'_f\boldsymbol{\beta}_f} < \frac{1}{s_f}(< 0). \tag{34}$$

Combining (33) and (34) yields $s_m < 1/s_f$. To prevent this from happening I need the reverse to hold, so $s_m > 1/s_f$. So if $s_m < 0$ and $s_f < 0$, a condition to ensure that the model does not generate the two outcomes $d_m = 1$, $d_f = 1$ and $d_m = 0$, $d_f = 0$ at the same time is $s_m s_f < 1$.

Next, note that the model outcomes $d_m = 1$, $d_f = 0$ and $d_m = 0$, $d_f = 1$ are not exclusive *a priori* either. The model generates these two outcomes, respectively if

$$\begin{aligned} \mathbf{X}'_m\boldsymbol{\beta}_m + s_m\mathbf{X}'_f\boldsymbol{\beta}_f &> 0,\\ \mathbf{X}'_f\boldsymbol{\beta}_f &\leq 0, \end{aligned} \tag{35}$$

and

$$\begin{aligned} \mathbf{X}'_m\boldsymbol{\beta}_m &\leq 0,\\ \mathbf{X}'_f\boldsymbol{\beta}_f + s_f\mathbf{X}'_m\boldsymbol{\beta}_m &> 0. \end{aligned} \tag{36}$$

Again, note that (35) and (36) cannot occur simultaneoulsy if either $s_m > 0$ or $s_f > 0$. If $s_m < 0$ and $s_f < 0$, combining (35) and (36) again yields the conditions (33) and (34). So again, I have $s_m s_f < 1$ if $s_m < 0$ and $s_f < 0$ to impose coherence.

Finally, reconsider (32). I may rewrite these conditions as

$$\begin{aligned} s_f(\mathbf{X}'_m\boldsymbol{\beta}_m + s^*_m\mathbf{X}'_f\boldsymbol{\beta}_f) &\leq 0,\\ s_m(\mathbf{X}'_f\boldsymbol{\beta}_f + s^*_f\mathbf{X}'_m\boldsymbol{\beta}_m) &\leq 0, \end{aligned} \tag{37}$$

with $s^*_m = 1/s_f$ and $s^*_f = 1/s_m$. First, assume that $s_m > 0$ and $s_f > 0$. Then (37) implies

$$\begin{aligned} \mathbf{X}'_m\boldsymbol{\beta}_m + s^*_m\mathbf{X}'_f\boldsymbol{\beta}_f &\leq 0,\\ \mathbf{X}'_f\boldsymbol{\beta}_f + s^*_f\mathbf{X}'_m\boldsymbol{\beta}_m &\leq 0. \end{aligned} \tag{38}$$

[29] For identification I need $s_m \neq 1/s_f$, but that is a different story from coherence.

This case does not imply incoherency, but I do see that there is a identification problem since parameters (s_m^*, s_f^*) will lead to the same probability as (s_m, s_f). Imposing $s_m s_f < 1$ for all positive values of s_m and s_f clearly excludes (s_m^*, s_f^*).

Next, assume that $s_m < 0$ and $s_f > 0$. Then (37) implies

$$\begin{aligned} \mathbf{X}_m'\boldsymbol{\beta}_m + s_m^*\mathbf{X}_f'\boldsymbol{\beta}_f &\leq 0, \\ \mathbf{X}_f'\boldsymbol{\beta}_f + s_f^*\mathbf{X}_m'\boldsymbol{\beta}_m &\geq 0. \end{aligned} \tag{39}$$

Thus, the outcome generated by the model is not unique. I have a combined coherence-identification problem here, which can easily be excluded by assuming $|s_m s_f| < 1$. The same holds for the case $s_m > 0$, $s_f < 0$. The final case $s_m < 0$, $s_f < 0$ has already been covered above.

Summarising, I have the following result: a sufficient condition for coherence of the model is $|s_m s_f| < 1$.

Note that it is not necessary to restrict either s_m or s_f to zero, as in the model by Heckman (1978) for limited dependent variables with endogenous right hand side variables, which in my notation reads

$$\begin{aligned} d_m^* &= \mathbf{X}_m'\boldsymbol{\beta}_m + (1 - d_f)s_m, \\ d_f^* &= \mathbf{X}_f'\boldsymbol{\beta}_f + (1 - d_m)s_f, \\ d_j &= \imath(d_j^* > 0). \end{aligned} \tag{40}$$

Using similar arguments as above, it is readily established that (40) requires $s_m s_f = 0$ for coherence, as was shown by Heckman (1978).

Table A1

Simulated Maximum Likelihood Estimates, Labour Supply Equations Parameters of Observed Characteristics

	Unmarried		Married	
Variable	Men	Women	Men	Women
log(age husb/17)	42.811**	4.621	59.777**	35.703**
	(8.051)	(7.585)	(9.736)	(9.188)
log(age husb/17) squared	−30.302**	−6.173	−49.004**	−17.329**
	(5.549)	(5.279)	(5.381)	(5.144)
log(age wife/17)	−27.154**	61.545**	−13.186	−3.339
	(6.603)	(7.165)	(8.447)	(7.911)
log(age wife/17) squared	14.507**	−49.510**	22.376**	−21.150**
	(4.916)	(4.691)	(4.945)	(4.706)
Educ level husb. 1	−1.971	−4.055**	−8.634**	−3.209**
	(2.203)	(1.443)	(0.900)	(0.893)
Educ level husb. 2	1.757*	−0.909	−4.005**	−1.164
	(1.012)	(1.010)	(0.765)	(0.718)
Educ level husb. 3	0.381	0.118	−3.176**	−0.706
	(0.774)	(0.750)	(0.561)	(0.573)
Educ level wife 1	3.551**	−8.120**	−0.555	−5.549**
	(1.135)	(1.503)	(0.853)	(0.857)
Educ level wife 2	1.324	−2.044*	1.579**	−2.907**
	(1.116)	(1.147)	(0.622)	(0.721)
Educ level wife 3	0.632	−2.808**	1.595**	−0.137
	(0.611)	(0.792)	(0.539)	(0.618)

** (*): significant at 5% (10%) level. Standard errors in parentheses.
Reference category of education: levels 4 and 5 (highest).

Table A2

Simulated Maximum Likelihood Estimates: The Wage Equations

Variable	Unmarried Men	Unmarried Women	Married Men	Married Women
intercept	2.591**	2.323**	3.052**	2.591**
	(0.059)	(0.054)	(0.053)	(0.049)
log(age/17)	0.675**	1.368**	−0.119	0.724**
	(0.157)	(0.136)	(0.129)	(0.121)
log(age/17) squared	0.053	−0.697**	0.419**	−0.364**
	(0.114)	(0.113)	(0.077)	(0.075)
Education level 1	−0.225**	−0.301**	−0.343**	−0.358**
	(0.059)	(0.055)	(0.025)	(0.031)
Education level 2	−0.303**	−0.244**	−0.360**	−0.338**
	(0.034)	(0.035)	(0.021)	(0.027)
Education level 3	−0.218**	−0.146**	−0.323**	−0.272**
	(0.029)	(0.030)	(0.018)	(0.027)
Education level 4	−0.092**	−0.038	−0.227**	−0.121**
	(0.028)	(0.031)	(0.019)	(0.026)
Technical	0.042*	0.062**	0.022*	−0.055*
	(0.023)	(0.026)	(0.012)	(0.033)
Econ./adm.	0.040*	0.060**	0.039**	0.042**
	(0.022)	(0.020)	(0.013)	(0.012)
General	−0.010	0.064**	0.041**	0.073**
	(0.022)	(0.021)	(0.016)	(0.013)
1992	0.046	0.014	0.017	0.051*
	(0.035)	(0.031)	(0.025)	(0.031)
1993	0.058*	0.082**	0.052**	0.074**
	(0.030)	(0.032)	(0.024)	(0.026)
1994	0.049*	0.027	0.019	0.088**
	(0.029)	(0.028)	(0.022)	(0.023)
1995	0.073**	0.074**	0.066**	0.097**
	(0.036)	(0.030)	(0.022)	(0.025)
1996	0.083**	0.115**	0.070**	0.122**
	(0.027)	(0.030)	(0.020)	(0.020)
1997	0.096**	0.123**	0.087**	0.152**
	(0.024)	(0.028)	(0.022)	(0.025)
1998	0.134**	0.176**	0.116**	0.193**
	(0.036)	(0.039)	(0.019)	(0.024)
1999	0.138**	0.167**	0.105**	0.246**
	(0.033)	(0.033)	(0.022)	(0.023)
2000	0.183**	0.201**	0.146**	0.251**
	(0.029)	(0.027)	(0.024)	(0.028)
2001	0.248**	0.332**	0.221**	0.364**
	(0.030)	(0.032)	(0.023)	(0.024)

** (*): significant at 5% (10%) level. Standard errors in parentheses.
Reference level of education: level 5 (highest); sector: services; year: 2002.

Table A3

Simulated Maximum Likelihood Estimates of the Covariance Matrices

Parameter	Unmarried	Married
σ_m	11.132**	10.715**
	(0.230)	(0.139)
σ_f	11.880**	12.164**
	(0.270)	(0.149)
σ_{mf}	−0.243**	−0.435**
	(0.024)	(0.015)
τ_m	0.244**	0.214**
	(0.003)	(0.002)
τ_f	0.247**	0.234**
	(0.004)	(0.002)
ρ_τ	0.162**	0.127**
	(0.024)	(0.018)
ρ_{mu}	−0.498**	−0.312**
	(0.030)	(0.026)
ρ_{fu}	−0.423**	−0.389**
	(0.042)	(0.020)
$\sigma_{\theta,m}$	8.761**	16.357**
	(0.275)	(0.240)
$\rho_{\theta,mf}$	0.233**	−0.393**
	(0.052)	(0.029)
$\sigma_{\theta,f}$	8.205**	10.147**
	(0.371)	(0.229)
$\sigma_{\omega,m}$	0.191**	0.220**
	(0.008)	(0.005)
$\rho_{\omega,mf}$	−0.177**	0.299**
	(0.076)	(0.028)
$\sigma_{\omega,f}$	0.134**	0.234**
	(0.010)	(0.007)
$\rho_{\theta\omega,m}$	−0.282**	−0.506**
	(0.041)	(0.026)
$\rho_{\theta\omega,fm}$	−0.307**	−0.660**
	(0.088)	(0.022)
$\rho_{\theta\omega,mf}$	0.064	0.037
	(0.048)	(0.031)
$\rho_{\theta\omega,f}$	−0.004	0.322**
	(0.091)	(0.031)

** (*): significant at 5% (10%) level. Standard errors in parentheses.

Table A4

Simulated Maximum Likelihood Estimates Labour Supply Equations: Dual Earner Couples Only

Variable	Unmarried Men	Unmarried Women	Married Men	Married Women
Intercept	−9.399	47.082**	17.018**	75.932**
	(7.152)	(5.742)	(5.487)	(5.233)
Wage husband	−0.606**	0.391	1.343**	1.635**
	(0.285)	(0.354)	(0.201)	(0.213)
Wage wife	1.048**	−0.704**	0.113	−1.039**
	(0.320)	(0.303)	(0.216)	(0.210)
$w_m/(w_m + w_f)$	70.498**	−33.411**	−6.037	−89.776**
	(12.424)	(12.136)	(9.121)	(9.542)
Wage husband squared/10	0.021	−0.043	−0.119**	−0.147**
	(0.038)	(0.048)	(0.022)	(0.023)
Wage wife squared/10	−0.088**	0.029	−0.018	0.086**
	(0.044)	(0.036)	(0.027)	(0.022)
Non-labour income/1000	−2.093	−2.947	−1.352	0.630
	(1.533)	(1.863)	(1.719)	(1.694)

** (*): significant at 5% (10%) level. Standard errors in parentheses.
The table only displays the coefficients of wages rates and non-labour income.

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